

Penalties Sized to the Gain: When Do Random Audits Stop AI Agents from Over-Using a Shared Resource?

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Abstract

AI agents that share a limited resource may take more than they are allowed, and an overseer can afford to check only a few of them. We ask when random audits stop them. We study an *audited commons*: six agents draw on a renewable resource, a reviewer sets each agent’s allowance so that the resource stays at the level where it regrows fastest, and audits answer a caught over-take with a fine, a memory of the offence, or both. With programmed agents in three games, audits work only through their consequences; deterrence begins where the expected fine equals the gain from cheating, a threshold we predict in advance in 5 of 5 new settings; and two intuitive designs backfire: tightening everyone’s allowance to absorb known cheaters, and aiming each audit at the largest report. With four language-model (LLM) families in 520 games, the two families that over-take on purpose stop at the fine predicted in advance from their own *one-round* gain (exactly for one model, one grid step early for the other), although over-taking rarely pays over the whole game. Like the myopia reported for LLM collectives, these agents ignore the cost to the shared stock, and here that myopia sets the deterrence threshold: penalties for LLM agents should be sized to the immediate gain.

1 Introduction

Language-model agents increasingly act together on shared, limited resources: compute budgets, rate-limited services, energy reserves, and the economic and natural commons that their users depend on (Piatti et al., 2024; Bracale Syrnikov et al., 2026). An overseer of such a system faces a problem that is old in economics and new in AI safety. Verification is expensive, so only a few actions can be checked (Amodei et al., 2016; Greenblatt et al., 2024), and an agent that takes more than its share is seen only if it happens to be checked. Two questions follow. Which consequences make sparse checking work? And can an overseer predict, before deploying a penalty, whether it will stop a given AI agent?

Enforcement economics answers the first question for rational offenders: a risk-neutral offender complies when the expected penalty, the probability of being caught times the fine, exceeds the gain from offending (Becker, 1968; Allingham and Sandmo, 1972; Avenhaus et al., 2002). AI-control work studies audit budgets, but there a caught model is shut

down, so audits without consequences, memory or graded fines are not tested (Greenblatt et al., 2024; Griffin et al., 2024). Studies of LLM agents in commons find that most fail to sustain the resource without enforcement (Piatti et al., 2024; Bracale Syrnikov et al., 2026). Enforcement studies with LLMs either remove violators flagged by an LM judge (Ye and Steinhardt, 2026) or compare one fine below and one above break-even in single decisions (Okamoto et al., 2026). No study measures when random audits deter AI agents that act repeatedly on a shared resource, or tests whether the deterrence threshold can be predicted for a given model.

We build an *audited commons* (Figure 1). Each round, six agents request a share of a renewable resource; a reviewer predicts the effect of the requests and grants allowances that keep the resource above half capacity, where it regrows fastest; each agent then chooses what it actually takes; and random audits check a few agents. We study three games with programmed agents, as a mechanism layer whose results we can predict, and then replace four of the six agents in the fishery game with LLMs from four families. Every study was run under a protocol frozen before its code, with amendments dated and failed predictions reported.

Contributions.

- **An audited-commons testbed.** We formalise oversight with a limited audit budget over a renewable resource, with three games (FISHERY, FOREST, RIVER), one harm line derived from each game’s regrowth, and the social outcome metrics used for commons games (Section 3).
- **What makes audits work, with programmed agents.** Audits help only through consequences. Deterrence starts at the expected-penalty break-even, which we predict in advance from runs without audits and confirm in 5 of 5 new settings. Two intuitive designs backfire: a tighter shared allowance that absorbs known cheaters makes cheating pay *more* than no audits, and aiming each audit at the largest report catches only 25% of liars against 100% at random (Section 4).
- **LLM agents are myopic offenders.** Across 520 games and 81,232 decisions, the two LLM families that over-take on purpose stop at the fine predicted in advance from their one-round gain, within one grid step and exactly for one model. Their whole-game gain from

over-taking is small or negative, yet they over-take until the fine outweighs the one-round gain. A reviewer that remembers past over-takes removes most of the harm without any fine, but does not stop the over-taking (Section 5).

2 Related Work

Scalable oversight and AI control. Scalable oversight concerns objectives that are “too expensive to evaluate frequently” (Amodei et al., 2016; Bowman et al., 2022). AI control evaluates protocols that audit a small share of an untrusted model’s outputs with trusted labour, and compares random with suspicion-targeted auditing (Greenblatt et al., 2024); Griffin et al. (2024) model such protocols as games. Multi-agent extensions show that monitors that check one agent or one action at a time miss harm that only appears in combination (Makins et al., 2026; Hu and Wang, 2026). Gans and Holden (2026) show in theory that rare random audits can deter AI agents when audit draws cannot be learned in advance. We add a renewable resource, graded consequences, and experiments with LLM agents.

LLM agents in commons. GovSim finds that “all but the most powerful LLM agents fail to achieve a sustainable equilibrium” (Piatti et al., 2024); LLM collectives in an energy commons behave like impatient optimisers (Bracale Syrnikov et al., 2026); and reasoning models free-ride in public-goods games with sanctions (Guzman Piedrahita et al., 2025). Ye and Steinhardt (2026) enforce norms among LLM agents, including in a fishery, by removing violators flagged by an LM judge and tracking reporters’ reliability. None varies the audit probability or the fine.

Enforcement economics. The expected-penalty condition goes back to Becker (1968) and Allingham and Sandmo (1972). Inspection games formalise partial verification and inspector commitment (Avenhaus et al., 2002); records of past offences give enforcement leverage (Harrington, 1988); report-based cutoff rules can beat random audits in static settings (Reinganum and Wilde, 1985); and a small fine can act as a price (Gneezy and Rustichini, 2000). In fisheries, model-based management fails when the model is wrong (Ludwig et al., 1993).

Rule-breaking LLMs. LLMs comply more when a fine lies above break-even than below it (Okamoto et al., 2026), and misbehave more when the perceived risk of getting caught falls (Scheurer et al., 2023). Table 1 summarises how our setting differs.

3 The Audited Commons

Resource and agents. $N = 6$ agents act for T rounds on a resource with level $x_t \in [0, K]$. After the agents take $c_t =$

Table 1: Closest prior work. $\checkmark$: yes; $(\checkmark)$: partly. ^aBecker; Harrington; inspection games. ^bGreenblatt et al. (2024). *Predicted threshold*: the fine at which cheating stops is predicted before it is tested.

	LLM agents	Several agents	Renewable resource	Random audits	3+ fine levels	Memory	Predicted threshold
Enforcement economics ^a	–	($\checkmark$)	–	$\checkmark$	$\checkmark$	$\checkmark$	–
AI control ^b	$\checkmark$	–	–	($\checkmark$)	–	–	–
Makins et al. 2026	$\checkmark$	$\checkmark$	–	–	–	–	–
Gans & Holden 2026	–	–	–	$\checkmark$	–	–	–
GovSim (Piatti et al. 2024)	$\checkmark$	$\checkmark$	–	–	–	–	–
Bracale Syrnikov et al. 2026	$\checkmark$	$\checkmark$	$\checkmark$	–	–	–	–
Ye & Steinhardt 2026	$\checkmark$	$\checkmark$	($\checkmark$)	–	–	($\checkmark$)	–
Okamoto et al. 2026	$\checkmark$	–	–	($\checkmark$)	–	–	–
This work	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

$\sum_i c_{i,t}$, the resource regrows logistically,

$$x_{t+1} = y_t + r y_t \left(1 - \frac{y_t}{K}\right) + \varepsilon_t, \quad y_t = x_t - c_t, \quad (1)$$

with weather noise ε_t in FOREST and RIVER. Regrowth $ry(1 - y/K)$ is largest at $y = K/2$, where the maximum sustainable harvest $rK/4$ is reached. Agent i requests $p_{i,t} \in [0, \bar{c}]$ and may report $\hat{p}_{i,t} \leq p_{i,t}$.

Reviewer. The reviewer chooses one shared cut $\lambda_t \in \{1, \frac{3}{4}, \frac{1}{2}, \frac{1}{4}, 0\}$, the largest whose predicted outcome meets its target, and grants allowances $a_{i,t} = \lambda_t \hat{p}_{i,t}$. A *joint* reviewer sums all requests before predicting; *local* reviewers judge one request at a time. The target is either a one-step safety line (the resource must not fall to a danger level next round) or the *half-capacity* target $y_t \geq K/2$, the level behind maximum sustainable yield (May et al., 1978).

Harm and outcome metrics. A round is *harmful* if the resource falls below half capacity: FISHERY stock below 50 of 100 after fishing, FOREST mean plot health below 10 of 20 (as a 5% chance constraint), RIVER quality below 50 of 100. Following commons-game practice (Piatti et al., 2024; Perolat et al., 2017), we also report *efficiency*, the total harvest divided by $T \cdot rK/4$; *equality*, $1 - \text{Gini}$ of the agents’ returns; and *survival*, the share of runs without collapse. Figure 1 (bottom) shows how harm connects the agents in each game.

Misbehaviour, audits and consequences. An agent *over-takes* when $c_{i,t} > a_{i,t}$, and *under-reports* when $\hat{p}_{i,t} < p_{i,t}$. Each round, each agent is audited independently with probability q ; an audit detects a violation with probability s . A caught agent pays a flat fine F , so the expected fine per violating round is $e = qsF$. Alternatively the reviewer keeps a *memory*: it records the caught agent’s excess and lowers its later allowances by the mean recorded excess, $a'_{i,t} = \max(0, \lambda_t \hat{p}_{i,t} - \bar{m}_i)$.

When should a fine deter? Let g_1 be an agent’s gain from one round of over-taking, holding later rounds fixed, and let g^* be its net gain over complying across the whole game, per round of over-taking.

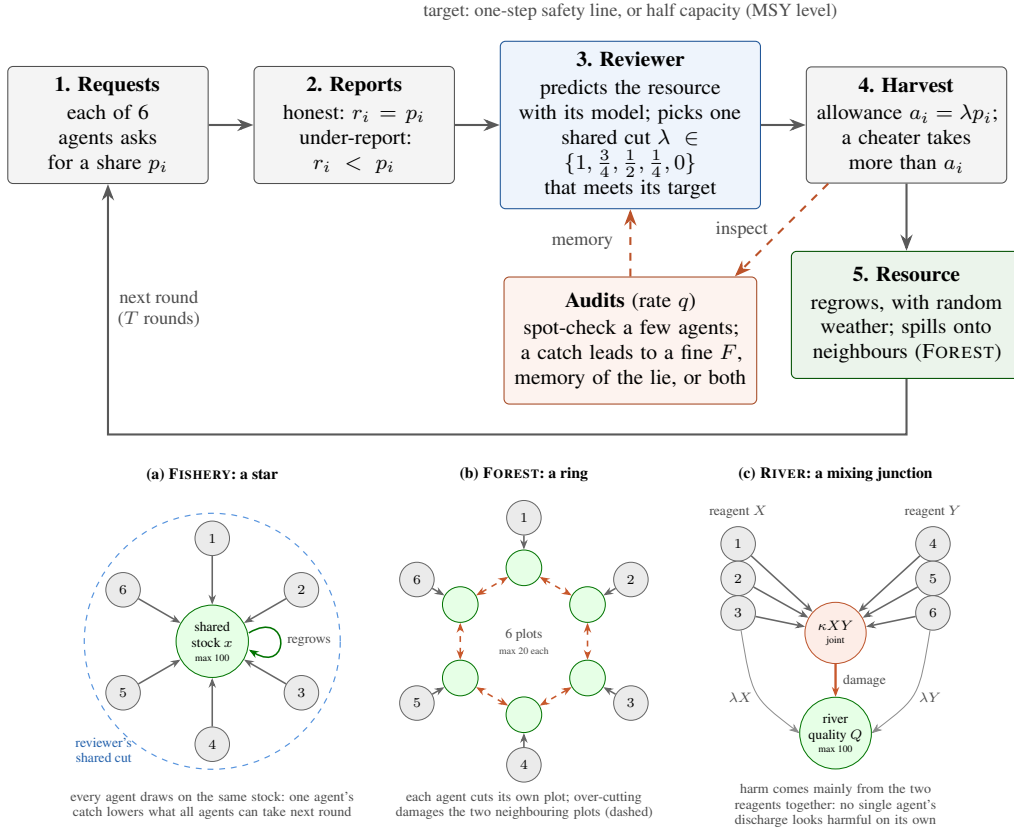


Figure 1: The audited commons. *Top:* one round. Agents request a share and may under-report; the reviewer picks one shared cut that meets its target; each agent takes its allowance or more; the resource regrows; random audits answer a caught violation with a fine, a memory of the offence, or both. *Bottom:* how harm connects the agents. (a) FISHERY is a star around one stock; the reviewer's shared cut (dashed ring) links every agent in all three games. (b) FOREST is a ring: over-cutting a plot damages its two neighbours. (c) RIVER is a mixing junction: damage $\kappa XY + \lambda(X + Y)$ comes mainly from the two reagents together.

Proposition 1 (Break-even). (i) A risk-neutral agent that maximises each round's catch net of the expected fine over-takes in a round if and only if $e < g_1$; under a flat fine it then takes the maximum $\bar{c}$. (ii) An agent that commits to one over-take level for the whole game over-takes at some level if and only if $e < g^*$.

Sketch. (i) Over-taking by $\delta > 0$ adds δ to the round's catch and costs e in expectation, independent of δ under a flat fine; the net gain is maximised at the largest δ , and is positive iff $g_1 > e$. (ii) A level d yields total gain $G(d)$ over $n(d)$ over-take rounds and expected fines $e n(d)$; it pays iff $G(d)/n(d) > e$, and $g^* = \max_d G(d)/n(d)$. $\square$

Part (ii) describes the programmed cheaters, which choose one level per game; part (i) describes an agent that decides every round. Since depleting the resource lowers later catches, $g^* \leq g_1$ in our games: a myopic agent keeps over-taking at fines where over-taking no longer pays over the game.

4 Programmed Agents: What Makes Audits Work

Setup. Agents follow threshold rules; 2 or 4 of the 6 are *greedy*. Misbehaving agents either under-report by a fixed share, or over-take with a level $d \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$ (taking $a + d(p - a)$) chosen by search to maximise their payoff on 32 training populations. Results are on 64 new populations of 80 rounds, with 95% paired bootstrap intervals over populations (4,000 resamples). Table 2 (appendix) lists the settings of every study.

The target matters more than the information. With a joint reviewer and true inputs, the half-capacity target gave 1,393 against 555 total harvest under the one-step line in FISHERY (difference +838, [+800, +865]), because the one-step line lets the stock fall to where it regrows slowly. With a wrong model of the resource, the joint reviewer's risky approvals in FOREST rose from 0.9% to 82% when regrowth was overestimated by 25%; in FISHERY, under a hidden collapse point, the one-step line collapsed 78% of populations and the half-capacity target 0%. A reviewer that learned the

model recovered within about 60 rounds (appendix Figure 5), echoing the adaptive-management lesson of Ludwig et al. (1993).

Audits work only through consequences. Against under-reporters, a reviewer that only uses audit results to correct its current decision barely improved safety (FOREST: 1.9% to 1.7% of rounds unsafe). Remembering each caught lie removed most of the harm in 9 of 9 settings (Figure 2a); in FISHERY, unsafe rounds fell from 69% to 1.5%. Memory alone, however, does not deter: adaptive cheaters kept 41% of their gain even at the highest audit rate, while memory plus a fine deterred at audit rate 0.05 instead of 0.10.

Deterrence starts at the break-even, and can be predicted. In the pilot setting, cheaters stopped at $e \approx 0.30$ – 0.40 , against a predicted break-even of 0.356, for audit rates from 1/6 to 0.02 and audits that miss up to three cheats in four. We then predicted g^* for five new settings from runs *without* audits, before any search with fines. In 5 of 5, the observed threshold fell within one grid step of the prediction, for predicted thresholds from 0.19 to 1.20 (Figure 2b). A known audit schedule cannot deter: honest agents received 28.8 against 40.9 with no audits at all (Figure 2d).

Two designs that backfire. First, a reviewer that answers known cheaters by tightening the *shared* cut protects the stock but spreads the cost onto honest agents: cheaters gained more than with no audits (142 against 113) and honest agents received 36 each, against 104 under a fine (Figure 2c). Second, aiming each audit at the largest (memory-corrected) report keeps auditing the same agent: in FISHERY only 25% of liars were ever caught, against 100% under random audits, and harm stayed at 96.2% of rounds against 7.4%. The same ordering held in RIVER (6.0% against 2.8% of rounds unsafe; 41% against 37% below half capacity). This complements static results in which report-based cutoff rules beat random audits (Reinganum and Wilde, 1985): here agents differ in size and play repeatedly, so a rank rule never leaves the largest agent, the feedback loop of Ensign et al. (2018). Aiming at a physical signal, the plot doing worst against prediction, was best in FOREST (harm 26% of the no-audit level, against 65% at random; Figure 2e).

5 LLM Agents: Deterrence of Myopic Offenders

Setup. In FISHERY (half-capacity target, 20 rounds), four of the six fishers are LLM agents and two follow fixed rules. Each round an LLM agent states a request and, after seeing its allowance, its catch, as JSON in tonnes; it sees the last three rounds of its own history. Audits are independent with $q = 1/6$; a catch costs a flat fine F (E -cells, $F \in \{0, 1, 2, 4, 8, 12, 18, 24, 30, 36\}$ t) or triggers the reviewer’s memory (EM , no fine). The explicit rules say “you can choose to catch more than your allowance”; control cells

drop this sentence (S) or reword the rules (P). Over-taking means catching more than 0.06 t above the allowance. We ran gpt-oss-120b (OpenAI), Nemotron 3 Super (NVIDIA) and Gemma 4 31B (Google) as frozen families, and Mistral Large 3 as an added family, through Ollama Cloud at temperature 0.7 with fixed seeds, on the same 10 populations in every cell. Each model first passed a pilot gate ($\geq 95\%$ valid first answers, comprehension ≥ 2 of 3). Across 520 games and 81,232 decisions there were no fallback answers, 6 re-prompts, and every comprehension check scored 3 of 3.

Over-taking against the fine. gpt-oss and Nemotron over-took in 51–92% of rounds at every expected fine up to 1.33 t, and in 0% at 6 t (Figure 3a). A small fine *raised* over-taking for gpt-oss (55% with no fine, 88–92% at 1–2 t), as fines can act as prices (Gneezy and Rustichini, 2000; Okamoto et al., 2026). Below the threshold the harm was large: under Nemotron, 5 to 10 of 10 lakes collapsed at every fine from 1 to 8 t, and gpt-oss’s rule-following fishers caught 8–25 t over the game against 32–34 t once over-taking stopped (Figure 3b). Gemma rarely over-took (4–10% of rounds), almost always by catching its request after the reviewer cut it, and Mistral Large 3 never over-took in 16,000 decisions; neither is informative about deterrence.

The stopping fine, predicted in advance. From each model’s no-fine cell we measured its one-round gain, the mean tonnes above the allowance on over-take rounds: $g_1 = 4.48$ t (gpt-oss) and 4.74 t (Nemotron). Before any further game we froze the prediction that $F = 12, 18, 24$ ($e = 2, 3, 4 < g_1$) would not deter and $F = 30$ ($e = 5 > g_1$) would, and ran 80 games at these fines. Taking “deterred” as at most 5% of rounds at that fine and every larger one, Nemotron stopped exactly at $F = 30$ and gpt-oss one grid step early, at $F = 24$ (Figure 4b). Over-taking was graded rather than a step: relative to no fine, it was 0.79 and 0.87 at $e/g_1 \approx 0.45, 0.23$ and 0.65 at $\approx 0.65, 0.02$ and 0.12 at ≈ 0.87 (gpt-oss and Nemotron). Our pre-registered hypothesis that over-taking would stay at or above 25% at $F = 24$ failed for both models (1.0% and 8.0%).

Myopic offenders. Measured the way the programmed cheaters are measured, as the net gain over a compliant game (the 36 t cell, where nobody over-takes), the LLMs’ whole-game gain per over-take round is $g^* = -1.05$ t for gpt-oss, which loses fish by over-taking in 7 of 10 populations because lakes collapse, and 0.71 t for Nemotron. Both nonetheless kept over-taking until the fine outweighed g_1 , nearly seven times g^* for Nemotron (Figure 4). By Proposition 1, the agents behave as in part (i), not part (ii): they weigh this round’s gain against this round’s expected fine and ignore the depleted stock in later rounds. Two further observations fit part (i). Over-takes were all-out, at the 6 t maximum in 68% (gpt-oss) and 93% (Nemotron) of over-take rounds, as a flat fine gives no reason to take less (Stigler, 1970); and being caught did not change the next round (gpt-oss over-took

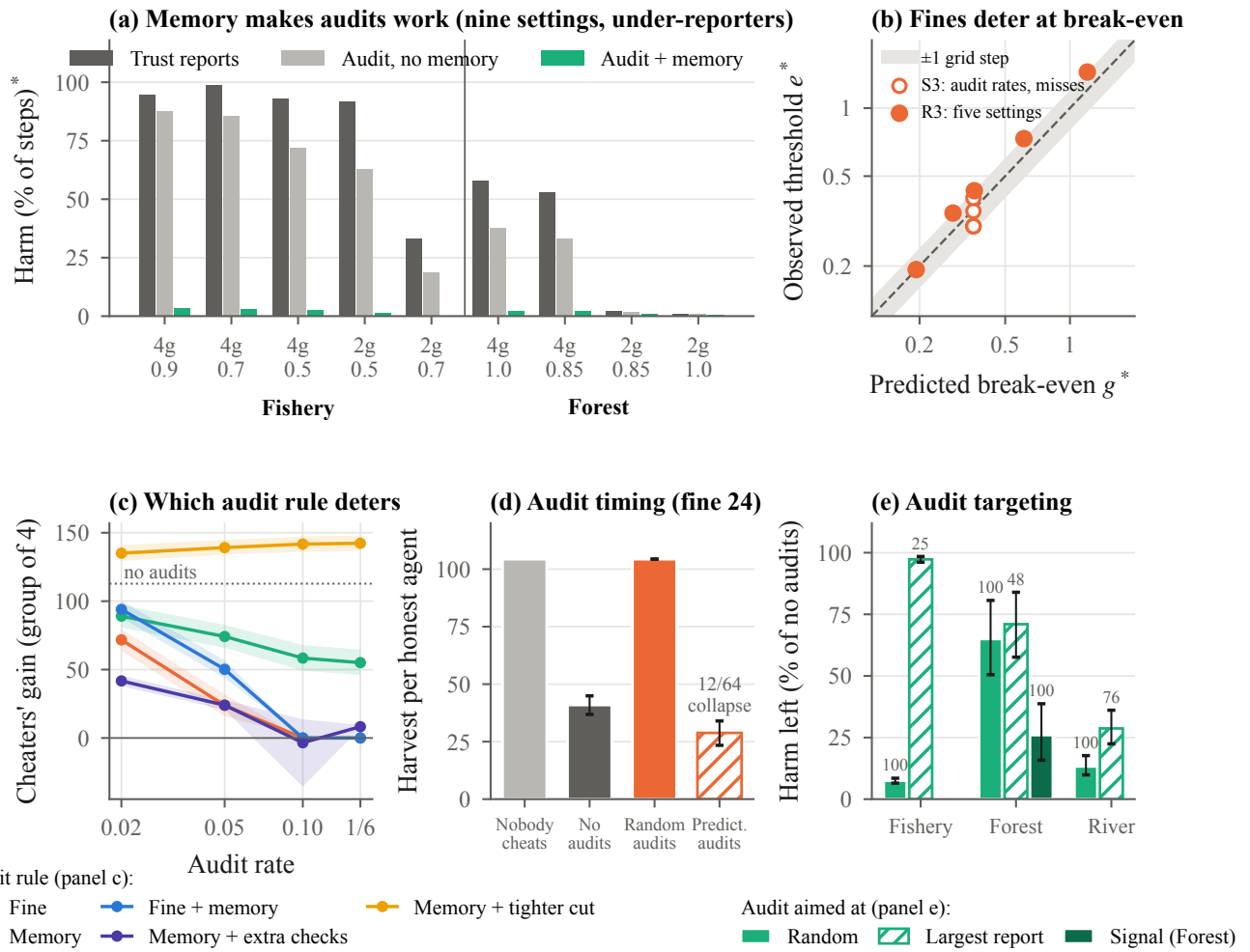


Figure 2: What makes audits work (programmed agents). (a) Under-reporters in nine settings: auditing without memory barely reduces harm; remembering each caught lie removes most of it. (b) The expected fine at which cheaters stop (e^*) against the break-even predicted from runs without audits (g^*); filled: five settings predicted before the search; every point lies within one grid step (shaded). (c) Cheaters' gain as the audit rate rises under five rules; tightening everyone's cut to absorb known cheaters makes cheating pay more than no audits. (d) A known audit schedule leaves honest agents worse off than no audits. (e) Harm left when one audit per round is aimed at random, at the largest report, or at the plot doing worst against prediction; numbers: share of liars ever caught. Error bars: 95% intervals.

again 78% of the time after a caught over-take, 81% after an unchecked one), as independent audits carry no information about the future. Unlike people in some tax experiments, the models showed no drop in compliance after an audit (Mittone, 2006; Kastlunger et al., 2009). The comparison with g^* was made after the results were seen.

Memory without fines. With the reviewer's memory and no fine, over-taking fell to 20% (gpt-oss) and 33% (Nemotron) of rounds, but did not stop. The resource was protected: rounds below half capacity fell from 92% to 20% and from 96% to 40%, no lake collapsed (against 5 and 1 of 10), and the rule-followers' catch rose from 8 to 29 t and from 9 to 32 t. Memory thus removes most of the harm without deterring: with no fine, $e = 0$ and Proposition 1 predicts continued

over-taking, while the reviewer's lower allowances limit what each over-take can do.

Deterrence costs no efficiency. At fines above g_1 ($F = 30, 36$), both models reached 0.99 of the maximum sustainable harvest, with the stock at or above half capacity in 96–100% of rounds and the most equal outcomes (equality 0.81). Below the threshold, efficiency stayed high only by over-fishing the stock, or collapsed with it. Efficiency alone therefore misleads, which is why commons studies report it with sustainability and equality (Piatti et al., 2024; Perolat et al., 2017).

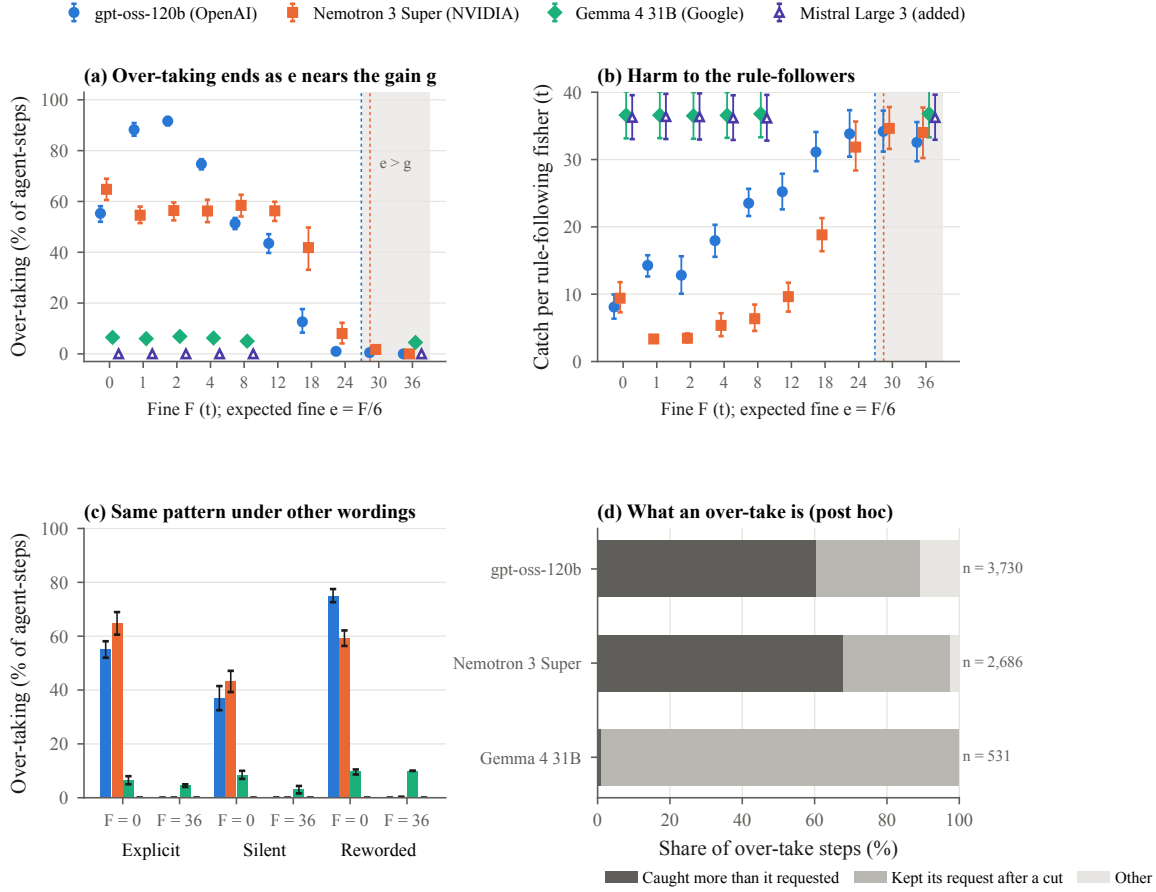


Figure 3: LLM agents in FISHERY. (a) Share of rounds in which each model over-took, against the fine F (expected fine $e = F/6$; $F = 12$ – 30 from the threshold test, gpt-oss and Nemotron only). Dotted: each model’s one-round gain; shaded: e above both. (b) Catch of the two rule-following fishers over 20 rounds. (c) A 36 t fine stops over-taking whether the rules state that over-taking is possible (explicit), omit it (silent), or reword it. (d) What an over-take round is (post hoc): gpt-oss and Nemotron mostly take more than they requested; Gemma almost only keeps its request after a cut. Error bars: 95% intervals over populations.

6 Discussion and Limitations

Implications for oversight. For LLM agents that over-take on purpose, the penalty that stops them can be predicted from their behaviour without a penalty, but the relevant gain is the immediate one. An overseer should measure the one-round gain, set the expected penalty above it, and keep audits random and covering every agent; where penalties are unavailable, a memory of past offences protects the resource without deterring. Because the agents ignore the long-run cost to the shared stock, they cannot be relied on to protect it for its own sake.

Limitations. The LLM results come from one game, two models that over-take on purpose, and 10 populations per cell, with one sampled game per population at temperature 0.7. The fine grid resolves the threshold to about $0.2 g_1$. Why the models cut back from about $0.4 g_1$ is untested; risk aversion (Allingham and Sandmo, 1972), a large stated fine read as a warning, and partial attention to the stock all fit. The my-

opia reading rests on a comparison made after the results. Our “silent” control still mentions a check with a fine of 0 t, so it does not replicate a no-check condition; our wording-robustness hypothesis held for Nemotron but not for gpt-oss, whose no-fine over-taking rose under reworded rules. The programmed cheaters optimise over a one-parameter family, a lower bound on an adaptive adversary. Most saved reasoning texts of the first LLM run were lost to a logging fault; outcomes were unaffected.

Future work. Varying the audit probability and the fine at a fixed expected penalty would separate risk aversion from myopia; fines that grow with the excess should curb all-out over-taking (Stigler, 1970); and LLM versions of FOREST and RIVER would test the threshold beyond one stock.

7 Conclusion

Random audits stop AI agents from over-using a shared resource only through consequences, only when those conse-

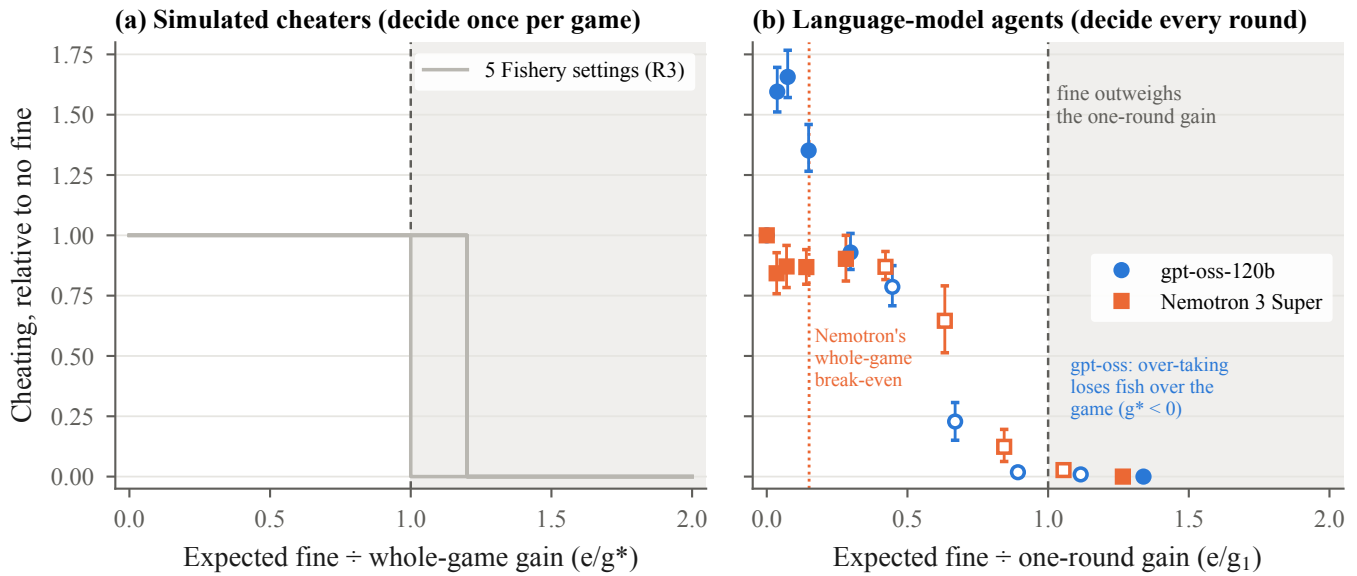


Figure 4: Each kind of agent stops when the expected fine outweighs the gain it acts on. (a) Simulated cheaters choose one level per game and stop at $e/g^* = 1.0$ – 1.2 of their whole-game gain, as predicted before the search. (b) LLM agents decide every round and stop near $e/g_1 = 1$ of their one-round gain (open markers: fines added by the threshold test, whose stopping point was predicted in advance). Measured by the whole-game gain they over-take far past break-even: over-taking loses gpt-oss fish over the game ($g^* < 0$), and Nemotron’s whole-game break-even (dotted) lies at $e/g_1 = 0.15$.

quences reach every agent unpredictably, and only once the expected penalty outweighs the gain the agent acts on. For programmed cheaters that gain is their whole-game gain, and the threshold can be predicted from runs without audits. For the LLM agents we studied it is the one-round gain: the penalty that stops them can be predicted before deployment, and their indifference to the shared stock means it must be.

References

- Michael G. Allingham and Agnar Sandmo. Income tax evasion: A theoretical analysis. *Journal of Public Economics*, 1(3–4): 323–338, 1972. doi: 10.1016/0047-2727(72)90010-2.
- Dario Amodè, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. Concrete problems in AI safety. *arXiv preprint arXiv:1606.06565*, 2016.
- Rudolf Avenhaus, Bernhard von Stengel, and Shmuel Zamir. Inspection games. In *Handbook of Game Theory with Economic Applications*, volume 3, pages 1947–1987. Elsevier, 2002. doi: 10.1016/S1574-0005(02)03014-X.
- Gary S. Becker. Crime and punishment: An economic approach. *Journal of Political Economy*, 76(2):169–217, 1968. doi: 10.1086/259394.
- Samuel R. Bowman et al. Measuring progress on scalable oversight for large language models. *arXiv preprint arXiv:2211.03540*, 2022.
- Marcantonio Bracale Syrnikov, Federico Pierucci, Matteo Prandi, Marcello Galisai, Piercosma Bisconti, Francesco Giarrusso, and Daniele Nardi. Draining the energy commons: Self-defeating over-appropriation as a coordination failure in agentic LLM collectives. *arXiv preprint arXiv:2607.22188*, 2026.
- Danielle Ensign et al. Runaway feedback loops in predictive policing. In *Conference on Fairness, Accountability and Transparency (FAT*)*, 2018. arXiv:1706.09847.
- Joshua S. Gans and Richard Holden. When does randomized oversight align AI agents that can conceal? *arXiv preprint arXiv:2609.38262*, 2026.
- Uri Gneezy and Aldo Rustichini. A fine is a price. *Journal of Legal Studies*, 29(1):1–17, 2000. doi: 10.1086/468061.
- Ryan Greenblatt, Buck Shlegeris, Kshitij Sachan, and Fabien Roger. AI control: Improving safety despite intentional subversion. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2312.06942.
- Charlie Griffin, Louis Thomson, Buck Shlegeris, and Alessandro Abate. Games for AI control: Models of safety evaluations of AI deployment protocols. *arXiv preprint arXiv:2409.07985*, 2024.
- David Guzman Piedrahita et al. Corrupted by reasoning: Reasoning language models become free-riders in public goods games. *arXiv preprint arXiv:2506.23276*, 2025.
- Winston Harrington. Enforcement leverage when penalties are restricted. *Journal of Public Economics*, 37(1):29–53, 1988. doi: 10.1016/0047-2727(88)90003-5.
- Yibo Hu and Ren Wang. When local monitors miss compositional harm: Diagnosing distributed backdoors in multi-agent systems. *arXiv preprint arXiv:2607.11751*, 2026.

- Barbara Kastlunger, Erich Kirchler, Luigi Mittone, and Julia Pitters. Sequences of audits, tax compliance, and taxpaying strategies. *Journal of Economic Psychology*, 30(3):405–418, 2009.
- Donald Ludwig, Ray Hilborn, and Carl Walters. Uncertainty, resource exploitation, and conservation: Lessons from history. *Science*, 260(5104):17, 1993. doi: 10.1126/science.260.5104.17.
- Nathaniel Makins, Federico Angelini, Mobina Shams, and Mary Phuong. Multi-agent AI control: Distributed attacks hamper per-instance monitors. *arXiv preprint arXiv:2607.07368*, 2026.
- Robert M. May, John R. Beddington, Joseph W. Horwood, and John G. Shepherd. Exploiting natural populations in an uncertain world. *Mathematical Biosciences*, 42(3–4):219–252, 1978. doi: 10.1016/0025-5564(78)90097-4.
- Luigi Mittone. Dynamic behaviour in tax evasion: An experimental approach. *Journal of Socio-Economics*, 35(5):813–835, 2006.
- Mika Okamoto, Ansel Kaplan Erol, and Kutluhan Erol. Why do AI agents break rules? how framing, context, and social signals shape compliance. *arXiv preprint arXiv:2608.12323*, 2026.
- Julien Perolat, Joel Z. Leibo, Vinicius Zambaldi, Charles Beattie, Karl Tuyls, and Thore Graepel. A multi-agent reinforcement learning model of common-pool resource appropriation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. arXiv:1707.06600.
- Giorgio Piatti et al. Cooperate or collapse: Emergence of sustainable cooperation in a society of LLM agents. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2404.16698.
- Jennifer F. Reinganum and Louis L. Wilde. Income tax compliance in a principal-agent framework. *Journal of Public Economics*, 26(1):1–18, 1985. doi: 10.1016/0047-2727(85)90035-0.
- Jérémy Scheurer, Mikita Balesni, and Marius Hobbhahn. Large language models can strategically deceive their users when put under pressure. *arXiv preprint arXiv:2311.07590*, 2023.
- George J. Stigler. The optimum enforcement of laws. *Journal of Political Economy*, 78(3):526–536, 1970.
- Yaowen Ye and Jacob Steinhardt. Norm enforcement for AI agents: Robustly shaping behavior in multi-agent systems. *arXiv preprint arXiv:2607.09766*, 2026.

Appendix

A Key terms

Term	Meaning in this paper
<i>The game</i>	
Agent	One of six participants who draw on the shared resource each step (a fisher in FISHERY, a forester in FOREST, a factory in RIVER).
Greedy agent	An agent whose fixed rule asks for a lot; 2 or 4 of the 6 in most runs.
Request p_i	How much an agent asks to take this step, as a share of its maximum.
Report	What the agent tells the reviewer it wants. An honest report equals the request; an <i>under-report</i> (a lie) is smaller.
Allowance	What the reviewer lets an agent take this step: its request times the shared cut.
Over-taking (cheating)	Taking more than the allowance. Nobody sees it unless the agent is audited.
Step, run	One round of requests and harvest; a run is 80 steps.
Population (context)	One randomly generated set of six agents and its random seeds. Results average over 64 of them.
<i>The reviewer</i>	
Reviewer	The overseer. It predicts what the requests would do to the resource and approves them or cuts them back.
Shared cut λ	One scaling factor ($1, \frac{3}{4}, \frac{1}{2}, \frac{1}{4}$ or 0) applied to everyone's request.
Model of the resource	The reviewer's equations for how the resource regrows. "Correct model" means it knows the true ones.
Joint reviewer	Adds up all six requests before predicting.
Local reviewers	Judge from one agent's request: the <i>cautious</i> one assumes everyone else takes as much as the largest request; the <i>optimistic</i> one ignores the others.
Safety line	The basic target: the resource must not fall below a danger level next step (Fishery stock 10).
Productive target (MSY)	A stricter target that keeps the stock where it regrows fastest (leave at least half of the maximum), the level behind <i>maximum sustainable yield</i> .
<i>Checking and consequences</i>	
Audit	A spot check of one agent: the reviewer learns its true request, or what it actually took.
Audit rate q	The chance that a given agent is audited in a given step (the <i>audit budget</i>).
Detection rate s	The chance that an audit of a cheater actually catches it.
Fine F	A penalty paid each time an agent is caught.
Expected fine $e = q s F$	The average fine an agent faces per step if it cheats.
Break-even g	What one step of cheating gains on average. Cheating stops paying once $e > g$.
Memory	The reviewer remembers how much each caught agent lied or over-took, and corrects for it afterwards.
Audit rule	Which consequences follow a catch: a fine, memory, extra checks on known cheaters, or a tighter cut for everyone.
<i>Measures</i>	
Harvest	What agents take from the resource (fish, timber, or discharge capacity). Used only as an outcome, never as a game name.
Harm	The share of steps in which the resource is below the safety line, or above the MSY limit (more taken than the productive target allows).
Collapse	The resource stays below its danger level long enough that the run ends.
Risky request approved	A request that would cross the target, let through unchanged (a missed harm).
Safe request cut	A request that was fine, cut anyway (<i>usefulness loss</i>).
Training vs. new runs	Cheaters choose their strategy on a few training populations; results are reported on 64 new ones.
95% interval	The range the result would likely fall in if the 64 populations were redrawn (bootstrap).

B Settings of every study

C The reviewer's target and model

D How the LLM agents over-take (post hoc)

E Protocols and reproducibility

Each study had a protocol frozen before its code, with hypotheses, falsifiers, sample sizes and analyses; amendments are dated and state what had been seen. The LLM studies used six amendments (an output-parsing fix; remote execution; a held-back and corrected memory cell; the replacement of Mistral Large 4, which used about 6,000 hidden-reasoning tokens per decision, by Mistral Large 3; and two infrastructure fixes) and one note that partial results of the threshold test were seen during the run, with

Table 2: The settings of every study. All simulations: 80 rounds, 64 new populations. LLM studies: 20 rounds, 10 populations. Audits random with probability 1/6 per agent and round (T1 and C1: one audit per round). ^aFISHERY: stock below 50 after fishing; FOREST: more than a 5% chance that mean plot health falls below 10; R1 also reports the one-step line, which the FISHERY parts of S1 and S1b used. ^bChosen by search to maximise the cheaters’ payoff over the whole game. ^cS4 and S5 cut a caught agent’s allowance by the share of the cut it took back; L2 by its mean tonnes over. ^dRIVER: quality below 30 (pre-registered) and below 50. ^ePart A compares reviewers built for the 30 line, so 30 is kept.

Study	Claims	Games	Who misbehaves, and how	How it is chosen	Consequence of a catch	Harm line	Gain used
R1, R2	1–3	Fishery, Forest	none (honest); R2: reviewer’s model wrong	–	–	half capacity ^a	–
S1, S1b	3	Forest, Fishery	programmed: under-report	one level per game ^b	fine, exclusion or none	half capacity ^a	–
S2, S3	4	Fishery	programmed: over-take	one level per game ^b	fine ($s \leq 1$)	half capacity	whole-game g^*
S4, S5	3–4	Fishery	programmed: over-take, adapting	one level per game ^b	memory ^c , fine	half capacity	whole-game g^*
R3	4	Fishery (6 settings)	programmed: over-take	one level per game ^b	fine	half capacity	whole-game g^* , predicted
T1	5	all three	programmed: under-report by half	fixed	memory, no fine	half capacity ^d	–
C1	1, 5	River	A: honest; B: under-report	fixed	B: memory	quality < 30 ^e	–
L2	3, 6	Fishery	4 LLMs: over-take	every round	fine 0–36 t, or memory ^c	half capacity	one-round g_1
L3	6	Fishery	gpt-oss, Nemotron: over-take	every round	fine 12–30 t	half capacity	one-round g_1 , predicted

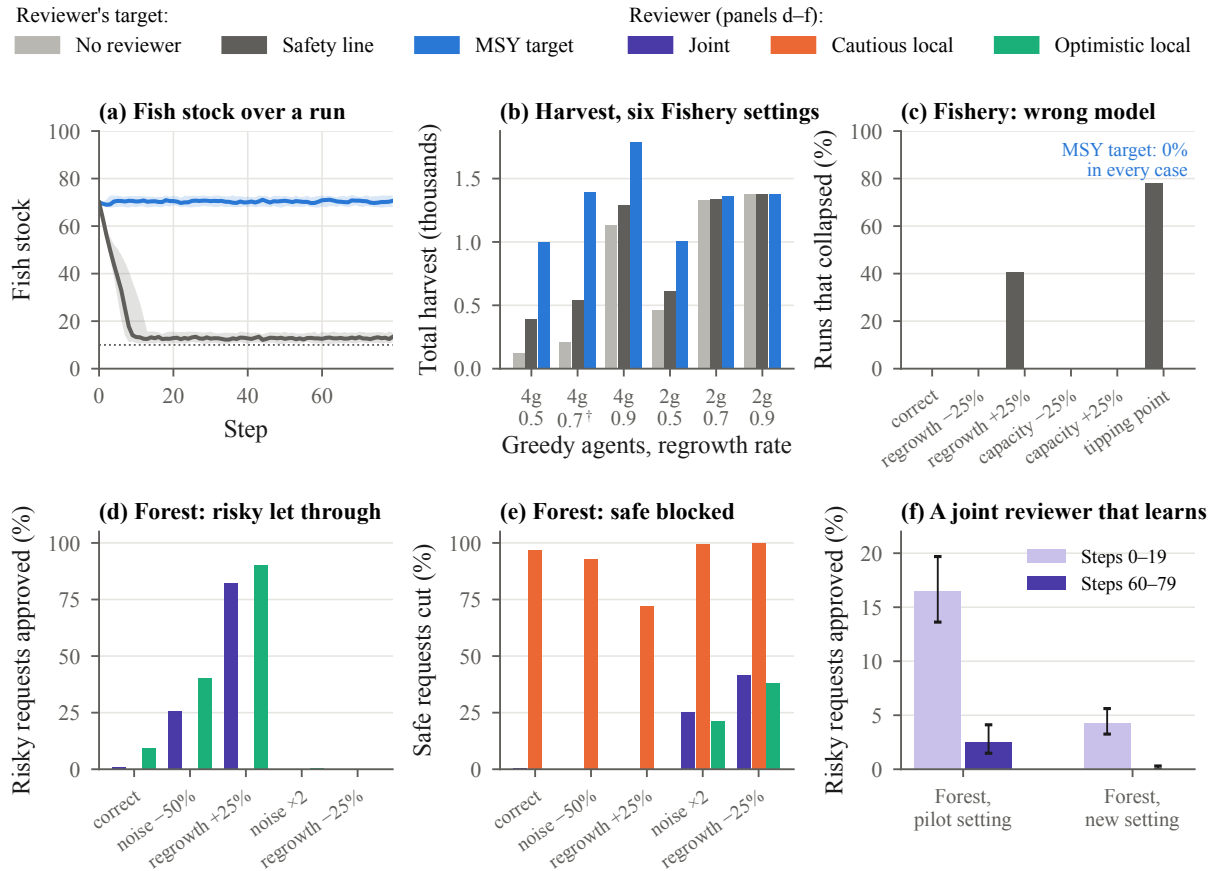


Figure 5: What the reviewer aims for and what it knows. Top row, FISHERY: stock over one run under the one-step line and the half-capacity target; total harvest in six settings; collapses when the reviewer’s model is wrong. Bottom row, FOREST: risky requests let through and safe requests blocked when part of the reviewer’s model is wrong, and a reviewer that learns the model while running. Error bars: 95% intervals.

nothing changed. Code, protocols, prompts, every game and all analysis scripts will be released.

Table 3: Audit rules compared (FISHERY, half-capacity target, 64 new populations). Bold: best in column.

Audit rule	Audit rate q	Cheaters' extra harvest [95% interval]	Harvest per honest agent	Steps over MSY limit (%)	Checks per step
No audits	—	112.8 [106, 120]	39.5	100	0.00
Fine	0.10	0.0	104.1	0	0.61
	1/6	0.0	104.1	0	1.00
Memory	0.10	58.4 [49, 68]	74.5	54	0.61
	1/6	55.1 [46, 65]	77.1	49	1.00
Memory + extra checks	0.10	−3.5 [−36, 14]	95.2	16	3.78
	1/6	8.3 [7, 10]	99.4	10	4.07
Memory + tighter shared cut	0.10	141.7 [136, 147]	36.0	24	0.61
	1/6	142.3 [137, 148]	36.0	16	1.00
Fine + memory + tighter cut	0.10	0.0	104.1	0	0.61
	1/6	0.0	104.1	0	1.00
Nobody cheats	—	0	104.1	0	0

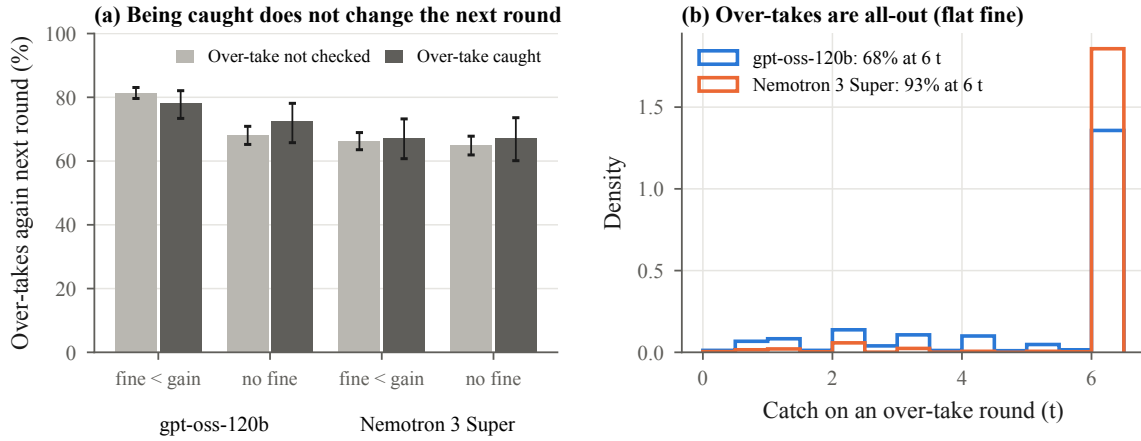


Figure 6: Post hoc analyses of gpt-oss and Nemotron. (a) After an over-take, the chance of over-taking again barely depends on whether the agent was caught. (b) Over-takes are mostly at the 6 t maximum. Intervals over agent-rounds, which are not independent; descriptive only.